

Requirements Specification Document

E-Commerce Order Management System

1. INTRODUCTION

This project is an E-Commerce Order Management System. It lets customers register, browse products, place orders, pay online, and track deliveries. It also lets sellers manage products and stock, and lets admins oversee the whole platform. The system is built on a database that stores customers, products, orders, payments, and reviews, and uses SQL to search, report, and analyze this data.

2. PROBLEM STATEMENT

Managing an online store by hand — tracking stock, orders, and payments in spreadsheets or separate systems — is slow and error-prone. This system brings everything into one connected database, so orders, stock, and payments always stay accurate and up to date.

3. OBJECTIVES

- Let customers register, browse, and buy products easily.
- Let sellers manage their products and stock.
- Process orders and payments securely.
- Track orders from placement to delivery.
- Generate simple reports (sales, ratings, inventory).

4. USER OF THE SYSTEM

User	What They Do
Customer	Buys products, tracks orders, leaves reviews
Seller	Adds products, manages stock, fulfills orders
Admin	Manages the platform, users, and reports

5. MAIN FEATURES

Customer Features:

- Register and log in to an account.
- Browse and search products by category, price, or keyword.
- View product details, price, stock, and reviews.
- Add products to a cart and check out.
- Pay online and receive an order confirmation.
- Track order status and view past orders.

- Leave a rating and review after delivery.

Seller Features:

- Add, edit, or remove product listings.
- Update stock levels and prices.
- View and fulfill incoming orders.

Admin Features:

- Manage product categories and approve sellers.
- View sales, inventory, and payment reports.
- Handle customer complaints or disputes.

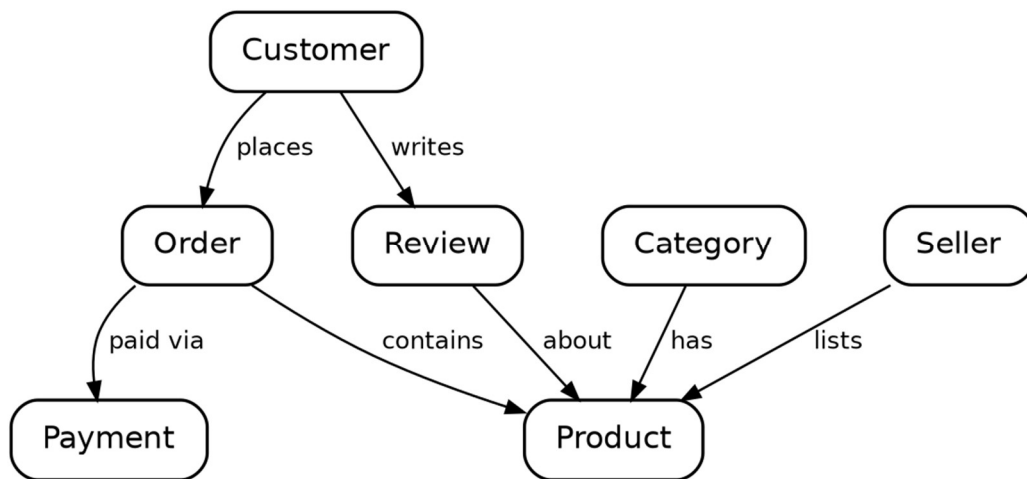
6. ORDER FLOWS

The diagram below shows the simple journey a customer takes, from logging in to leaving a review.



7. RELATIONSHIP

The diagram below shows, in simple terms, how the main pieces of information relate to each other in the database.



8. BASIC REQUIREMENTS

Type	Requirement
Performance	Pages and searches should respond quickly, even with many users.
Security	Customer and payment data must be protected and access-controlled.
Reliability	Orders and payments must always be recorded correctly, with no data loss.
Scalability	The system should handle growth in products, sellers, and orders.
Usability	The system should be simple, clear, and easy to use.

9. INPUTS AND OUTPUTS

Inputs:

- Customer details — name, email, password, address.
- Product details — name, category, price, stock.
- Order details — products, quantity, delivery address.
- Payment details — amount, payment method.

Outputs:

- Order confirmation and invoice.
- Order status and delivery updates.
- Sales, inventory, and rating reports.